RNDS-B-00008 v2.0
36-02-026 --D
25953NDS04 [4]

RENAULT NISSAN DESIGN SPECIFICATION
(RNDS)
Vehicle generic specifications
Design specification
RNDS-B-00008 v2.0

RENAULT STANDARD (GDNormes) 36-02-026 --D	NISSAN DESIGN SPECIFICATION (NDS) 25953NDS04 [4]

Issued: 2017-03-31

IMPORTANT PART SYMBOL
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Title: CAN High-Speed and CAN-FD Hardware

Confidentiality level: Renault Nissan Internal

Signature (Alliance Single Technical Leader)

T. De Soete

Signature (Renault validator)

Signature (Nissan validator)

Revisions

S. LOUVART T. MATSUMOTO	V. BIDAULT T. MOGARI	DEA-SIA / XX2	2017.03.31	Typography Mistake Corrections	5.0	--D	4
S. LOUVART T. MATSUMOTO	V. BIDAULT T. MOGARI	DEA-SIA / XX2	2016.03.31	New Template Update for CAN-FD Clarification of Clock Tolerance requirements	4.0	--C	3
S. LOUVART T. MATSUMOTO	N. MOLL K. NATORI	65610 / XX6	2010.03.01	Test Plan in a separate document Corrections and update Cin max and CMC LMin updated Resonator acceptance clarified	3.0	--B	2
L. ROY T. MATSUMOTO		65610 / KP1	2004.04.01	Change name to CAN HS Hardware Corrections and update	2.0	--A	1
	P. ARTUR	65610 / KR3	2001.07.01	Newly established	1.0	---	N
APPROVED BY	SIGN	SECT	APPROVED DATE	ALTERATION	CHG RNDS	Ver GDNormes	CHG NES

RENAULT-NISSAN

Important Notices and Disclaimers

- "This document is identical to NISSAN document No. 25953NDS04
  It complies with the agreement reached between RENAULT and NISSAN in 2017-03-31.
  Any revision or alteration of this document is subject to prior approval by the RENAULT (or NISSAN) secretariat.
- This document was translated from the English original. In case of any discrepancies, the English original shall take precedence over any other language translation.

FOREWORD

No information

Renault	Issued by	:	V. BIDAULT
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	Validated by	:	T. MATSUMOTO

Contents

INTRODUCTION

No information

PRODUCT SPECIFICATION

1. SCOPE

This document describes the requirements for the CAN Hardware interface of an ECU. It is applicable for CAN-HS and CAN-FD ECU's. Each requirement is described with a requirement number and a description in a chart such as presented below:

[CANHWXXXy]: description of the requirement; where XXX represents the requirement number for the whole document, and y represents the revision information of the requirement. This requirement is applicable for a CAN-HS or a CAN-FD ECU.

[CANHW-HSXXXy]: description of the requirement; where XXX represents the requirement number for the whole document, and y represents the revision information of the requirement. This requirement is applicable only for a CAN-HS ECU.

[CANHW-FDXXXy]: description of the requirement; where XXX represents the requirement number for the whole document, and y represents the revision information of the requirement. This requirement is applicable only for a CAN-FD ECU.

Proofs of the respect of all requirements are requested. A validation test plan for all requirements is available in [REF 1] and shall be used to prove that all requirements are fully respected.

2. NORMATIVE REFERENCES

These normative references apply the latest version.

[REF 1]	Validation Test Plan for CAN Hardware Interface: RENAULT:36-02-016 / NISSAN 25953NDS47
[REF 2]	CAN Sleep Management: RENAULT: 36-02-027 / NISSAN: 25953NDS05
[REF 3]	EMC requirements: RENAULT: 36-00-808 / NISSAN: 28401NDS02 and ECU specific requirements
[REF 4]	Specification of CAN Message Set for specific ECU.
[REF 5]	Component list: RENAULT: NT65610-2005-129 / NISSAN: 25953NDS37
[REF 6]	Result Report For CAN Physical Interface: RENAULT-NISSAN: NT65610-2003-180
[REF 7]	ISO 11898 Part 2 (Edition of 2015)
[REF 8]	CRS (Communication Requirement Specification)

3. TERMS AND DEFINITION

For the purpose of this standard, the following terms and definitions apply:

- **Arbitration phase:** phase where the nominal bit time is used.
- **Data phase:** phase where the data bit time is used.
- **Bus Driver:** electronic circuit converting information or data signals into physical signals so that these signals can be transferred across the communication medium.
- **Transceiver:** electronic circuit that connects a CAN node to a CAN bus, consisting of a bus comparator and a bus driver.

4. SYMBOLS AND ABBREVIATED TERMS

For the purpose of this standard, the following symbols and abbreviated terms apply.

Acronym/Other	Meaning
BRP	**B**aud **R**ate **P**rescaler
BTL	**B**it **T**ime **L**ogic
CAN	**C**ontroller **A**rea **N**etwork
CAN_HS	CAN **H**igh-**S**peed.
CAN_FD	CAN with **F**lexible **D**atarate
DC	**D**irect **C**urrent
ECU	**E**lectronic **C**ontrol **U**nit
EMC	**E**lectro-**M**agnetic **C**ompatibility
EUT	**E**quipment **U**nder **T**est
ID	**ID**entifier
PCB	**P**rinted **C**ircuit **B**oard
SJW	re**S**ynchronization **J**ump **W**idth
TTL	**T**ransistor-**T**ransistor **L**ogic
Tq	**T**ime **q**uantum

5. COMMUNICATION CIRCUIT SPECIFICATIONS

5.1. COMMUNICATION INTERFACE CIRCUIT CONFIGURATION

[CANHW001a]: The configuration of the communication interface circuit shall be as shown in Figure 5-1.

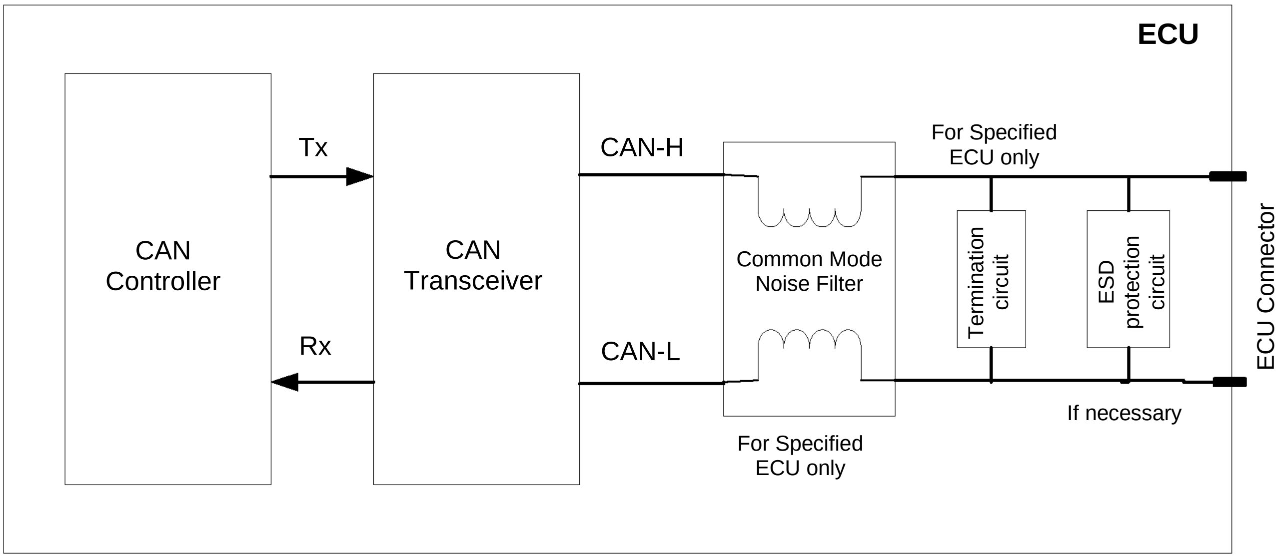


Figure 5-1: CONFIGURATION OF COMMUNICATION INTERFACE CIRCUIT

IMPORTANT REMARKS: In order to enhance EMC performances of the ECU, it is recommended:

- To place the CAN hardware interface (composed of all components between Tx, Rx and ECU connector) as close as possible to the ECU connector,
- To place the EMC or ESD protections, if they are necessary, between the ECU connector and the CAN transceiver. Those devices have capacitance: if they are added, requirement in **Table 5-4** shall be satisfied,
- To design the PCB in order to route CAN_H and CAN_L as close as possible to each other and as symmetrical as possible.

[CANHW002a]: The PCB layout shall be designed to allow the mount of Termination circuit impedance according to chapter 5.3. without any redesign of the ECU (meaning that the PCB layout provisions for Termination circuit impedance is mandatory).

[CANHW-HS003a]: The PCB layout shall be designed to allow the mount of Common Mode Noise Filter according to chapter 5.4. without any redesign of the ECU (meaning that the PCB layout provisions for the Common Mode Noise Filter is mandatory).

[CANHW-FD004a]: The Common Mode Noise Filter shall be mounted on the ECU according to chapter 5.4.

[CANHW005a]: When providing a Termination Circuit and a Common Mode Noise Filter, the Termination Circuit shall be placed on the outside (on the side of communication bus) of the Common Mode Noise Filter.

[CANHW-HS006a]: Transceiver used in the ECU shall be chosen in the list defined in [REF 5], §4.1 or §4.2

[CANHW-FD007a]: Transceiver used in the ECU shall be chosen in the list defined in [REF 5], §4.2.

[CANHW008a]: **The CAN interface schematic** shall be sent back with [REF 6], including all components and references (resistor, capacitor, filter, varistor, and so on...) connected to:

- Any pins of the CAN transceiver,

– The CAN_H and CAN_L lines.

5.2. ELECTRICAL CHARACTERISTICS OF COMMUNICATION INTERFACE CIRCUIT

In conformity with [REF 7], the communication interface circuit shall meet the requirements described in the following sections (1) through (8) at the operating voltage and operating temperatures including fluctuations caused by deterioration with ageing.

The operating voltage and operating temperatures shall conform to the specific ECU specifications.

(1) ECU SINGLE UNIT DC OUTPUT CHARACTERISTICS

[CANHW009a]: When disconnected from the communication bus, the output of the ECU shall be within the range specified in Table 5-1.

Table 5-1: ECU SINGLE UNIT OUTPUT DC CHARACTERISTICS

Signal	Items	Symbol	Unit	Lower limit	Standard	Upper limit	Condition
Dominant	Output bus voltage	V_{CAN_H}	V	2.75	3.5	4.5	60 Ω between terminals CAN-H ~ CAN-L
		V_{CAN_L}	V	0.5	1.5	2.25	
	Differential output bus voltage	V_{DIFF}	V	1.5	2.0	3.0	
Recessive	Output bus voltage	V_{CAN_H}	V	2.0	2.5	3.0	No load between terminals CAN-H ~ CAN-L
		V_{CAN_L}	V	2.0	2.5	3.0	
	Differential output bus voltage	V_{DIFF}	mV	- 500	0	50	

(2) SINGLE UNIT DC INPUT CHARACTERISTICS

[CANHW0010a]: When a signal specified in Table 5-2 is transmitted to the communication interface circuit with the communication bus disconnected, the ECU shall detect signal level.

Table 5-2: ECU SINGLE UNIT INPUT DC CHARACTERISTICS

Signal	Items	Symbol	Unit	Lower limit	Standard	Upper limit	Condition
Dominant State	Differential input voltage	V_{DIFF}	V	0.9	-	5.0	60 Ω between terminals CAN-H ~ CAN-L
Recessive State	Differential input voltage	V_{DIFF}	V	- 1.0	-	0.5	

(3) BUS POTENTIAL REQUIRED FOR COMMUNICATION IN THE COMMON MODE

[CANHW0011a]: When an output signal is transmitted with the ECU connected to the communication bus, the potential of the communication bus shall meet the requirements in Table 5-3.

Table 5-3: POTENTIAL OF COMMUNICATION BUS IN THE COMMON MODE

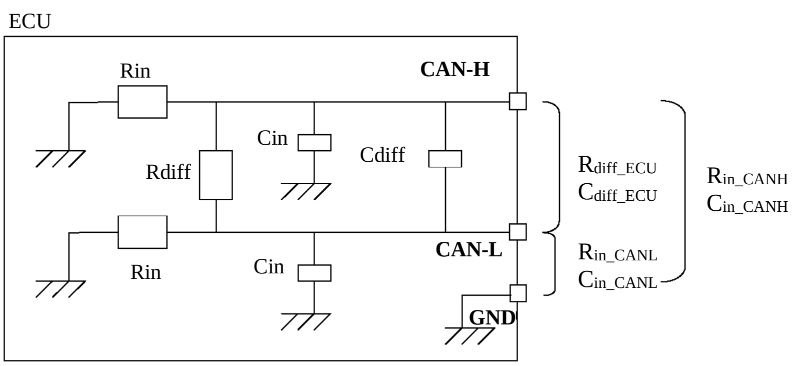
Signal	Items	Symbol	Unit	Lower limit	Standard	Upper limit	Condition
Dominant	Common mode bus voltage	V_{CAN_H}	V	- 10.8	3.5	12.0	Measuring using the grounds of each ECU
		V_{CAN_L}	V	- 12.0	1.5	10.8	
	Differential bus voltage	V_{DIFF}	V	1.2	2.0	3.0	Measure using all ECUs
Recessive	Common mode bus voltage	V_{CAN_H}	V	-12.0	2.5	12.0	Measure using the grounds of each ECU
		V_{CAN_L}	V	-12.0	2.5	12.0	
	Differential bus voltage	V_{DIFF}	V	- 0.12	0	0.012	Measure using all ECUs

(4) ECU IMPEDANCE AND CAPACITANCE

[CANHW0012a]: When disconnected from the communication bus, the impedance and capacitance of the ECU shall be within the range specified in Table 5-4.

Table 5-4: ECU IMPEDANCE AND CAPACITANCE

Signal	Items	Symbol	Unit	Lower limit	Standard	Upper limit	Condition
Recessive	Internal resistance	R_{in_CANH} R_{in_CANL}	kΩ	6	-	50	See Note for R_{in} complete calculation.
	Differential internal resistance	R_{diff_ECU}	kΩ	12	-	100	
	Internal capacitance	C_{in_CANH} C_{in_CANL}	pF	-	20	50	See Note for C_{in} complete calculation.
-	Differential internal capacitance	C_{diff_ECU}	pF	-	10	-	



$C_{diff\_ECU} = C_{diff} + C_{in}/2$

C_{in_CANX} and C_{diff_ECU} are not only additional Capacitances.
They include the following elements:
- the input capacitance of Transceiver
- the capacitance of additional device on CAN line
- the capacitance of printed wiring board

ECU EQUIVALENT CIRCUIT

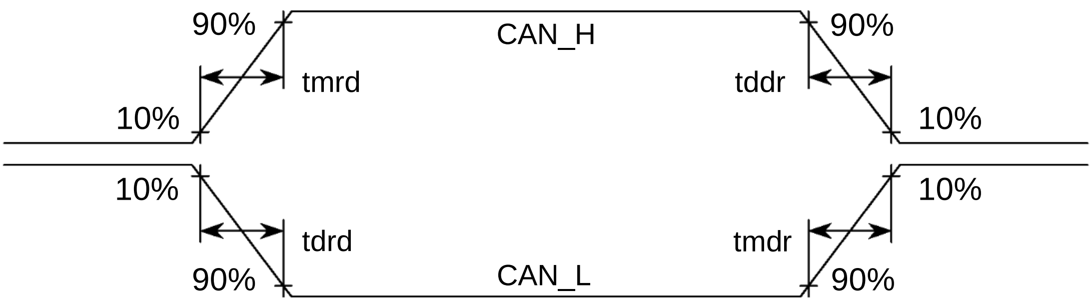
Note: The circuit configuration without the additional capacitance on the CAN_H and CAN_L line is recommended. However, if additional components are necessary because of EMC, internal capacitance C_{in_CANH} and C_{in_CANL} shall be less than 50pF.

(5) DRIVER OUTPUT EDGE SLOPE

[CANHW0013a]: The recessive to dominant transition and dominant to recessive transition on CAN bus shall meet the requirements in Table 5-5.

Table 5-5: DRIVER OUTPUT EDGE SLOPE

Items	Symbol	Unit	Lower limit	Standard	Upper limit
Recessive to dominant transition on CAN_H	t_{mrd}	ns	-	-	100
Recessive to dominant transition on CAN_L	t_{drd}				
Dominant to recessive transition on CAN_H	t_{ddr}				
Dominant to recessive transition on CAN_L	t_{mdr}				



IMPORTANT REMARK

No lower limit of transition time is requested, though transitions too fast could cause EMC problems, such as emitted and conducted radiations.
However, the ECU has to conform to EMC requirements [REF 3].

(6) DRIVER VOLTAGE SYMMETRY

[CANHW0014a]: The Voltage Symmetry of the ECU shall be within the range specified in Table 5-6.

Table 5-6: Driver Voltage Symmetry

Signal	Items	Symbol	Unit	Lower limit	Standard	Upper limit	Condition
Recessive, Dominant, or Transition	Voltage Symmetry	V_{sym}	-	0.9	1.0	1.1	$V_{sym} = (V_{CAN\_H} + V_{CAN\_L}) / V_{cc}$ 60Ω between terminals CAN-H and CAN-L.

Note: T_{XD} of the transceiver is stimulated by a square wave signal with a frequency that corresponds to the highest bit rate for which the application is intended. For instance a frequency of 1MHz corresponds to a maximum data rate of 2Mb/s. V_{cc} is the supply voltage of the transceiver.

(7) DRIVER OUTPUT BIT TIMING SYMMETRY

For data communication upper than 1Mbps, asymmetry of Bit Timing can occur, recessive bits being shorter than dominant bits. Worst case is obtained after a transition of 5 Dominant Bits. Data-rate selection for Transceiver selection shall be made according to each ECU specification ([REF 8])

[CANHW-FD015a]: The recessive bit duration after a 5 dominant bit transition (worst case) on CAN bus for a 2Mbps capable transceiver shall meet the requirements in Table 5-7.

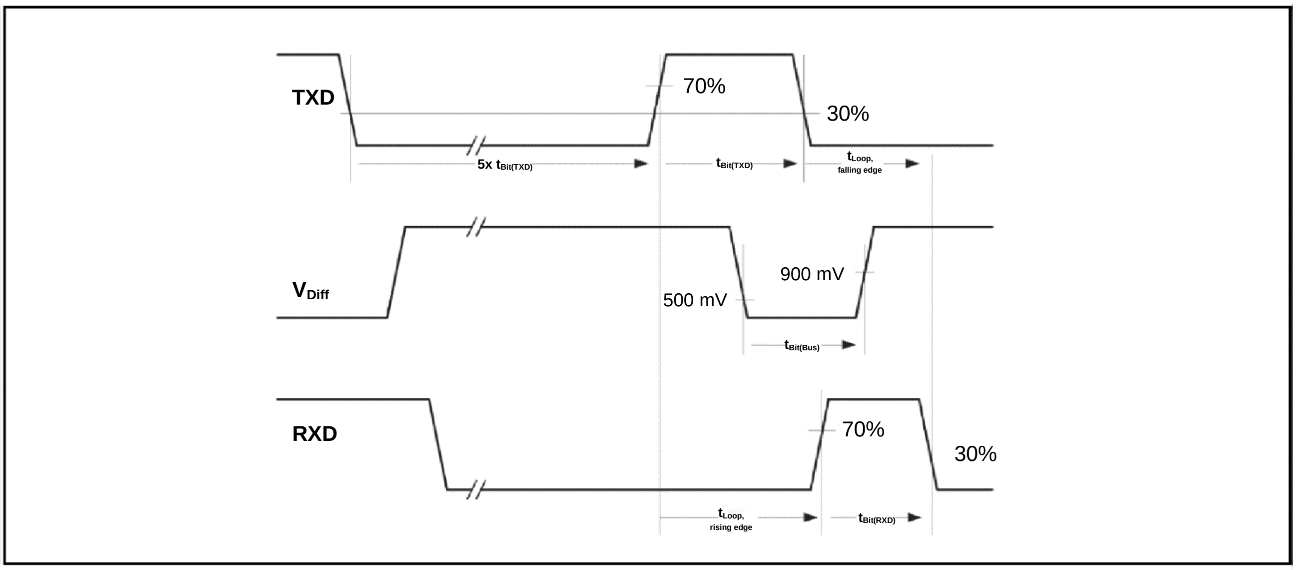
Table 5-7: DRIVER OUTPUT BIT TIMING SYMMETRY (for 2Mbps transceiver)

Items	Symbol	Unit	Lower limit	Standard	Upper limit
Transmitted Recessive Bit Duration on VDiff	$t_{Bit(Bus)}$	ns	435	500	530
Received Recessive Bit Duration on Rxd	$t_{Bit(RXD)}$		400		550
Receiver Timing Asymmetry	$t_{Bit(RXD)}$- $t_{Bit(Bus)}$	ns	-65	0	40

[CANHW-FD016a]: The recessive bit duration after a 5 dominant bit transition (worst case) on CAN bus for a 5Mbps capable transceiver shall meet the requirements in Table 5-8.

Table 5-8: DRIVER OUTPUT BIT TIMING SYMMETRY (for 5Mbps transceiver)

Items	Symbol	Unit	Lower limit	Standard	Upper limit
Transmitted Recessive Bit Duration on VDiff	$t_{Bit(Bus)}$	ns	155	200	210
Received Recessive Bit Duration on Rxd	$t_{Bit(RXD)}$		120		220
Receiver Timing Asymmetry	$t_{Bit(RXD)}$- $t_{Bit(Bus)}$	ns	-45	0	15

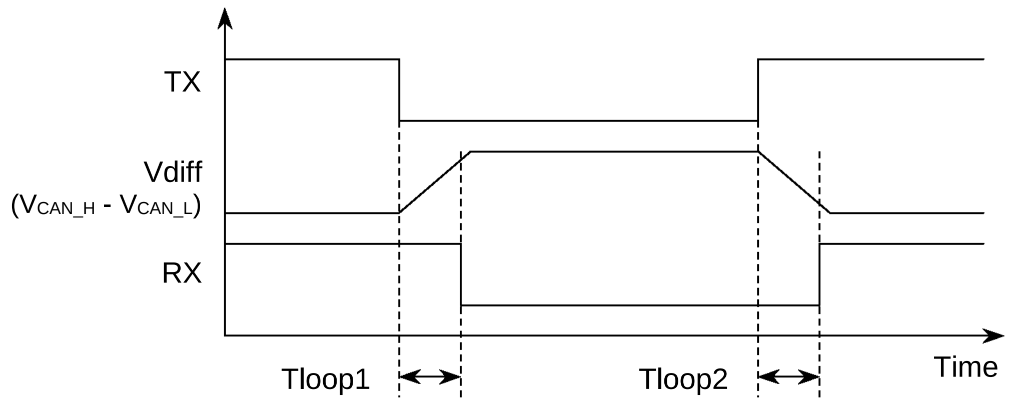


(8) CAN TRANSCEIVER DELAY TIME

[CANHW0017a]: The transceiver delay time (tloop1 and tloop2) shall meet the requirements in Table 5-9.

Table 5-9: CAN TRANSCEIVER DELAY TIME

Items	Symbol	Unit	Lower limit	Standard	Upper limit
Recessive to dominant delay between Tx and Rx	tloop1	ns	-	-	255
Dominant to recessive delay between Tx and Rx	tloop2				



5.3. TERMINATION CIRCUIT

To determine the termination circuit applicability, refer to specification requirements provided separately for each system.

[CANHW018a]: The main termination circuit of Type 1 shall be designed according to Figure 5-2 and values from Table 5-10.

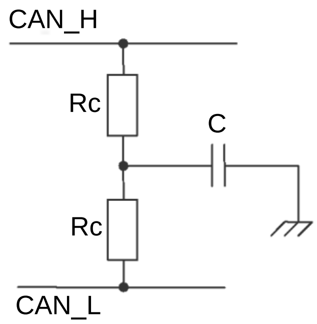


Figure 5-2: TERMINATION CIRCUIT TYPE 1

Table 5-10: VALUES FOR MAIN TERMINATION CIRCUIT TYPE 1

Items		Symbol	Unit	Lower limit	Standard	Upper limit	Remarks
Termination resistance	Nominal value	Rc	Ω	60	-	65	Maximum deviation: ± 1 % Minimum power consumption: 110 mW
Termination capacitance	Nominal value	C	nF	-	4.7	-	Maximum deviation: ± 10 % Minimum Voltage 50V

[CANHW019a]: The use of SPLIT pin of a transceiver is forbidden.

[CANHW020a]: The stub termination circuit of Type 1 shall be designed according to Figure 5-2 and values from Table 5-11.

Table 5-11: VALUE FOR STUB TERMINATION CIRCUIT TYPE 1

Items		Symbol	Unit	Lower limit	Standard	Upper limit	Remarks
Termination resistance	Nominal value	Rc	Ω	-	2400	-	Maximum deviation: ± 1 % Minimum power consumption: 110 mW
Termination capacitance	Nominal value	C	nF	-	0.22	-	Maximum deviation: ± 10 % Minimum Voltage 50V

[CANHW021a]: The main termination circuit of Type 2 shall be designed according to Figure 5-3 and values from Table **5-12**.

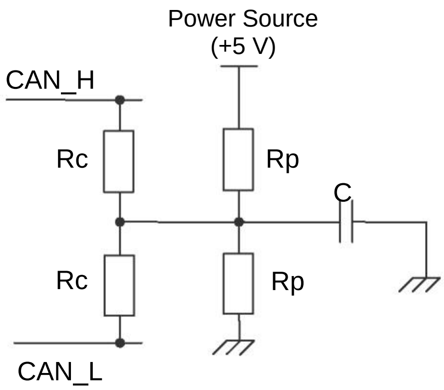


Figure 5-3: TERMINATION CIRCUIT TYPE 2

Table 5-12: VALUES FOR TERMINATION CIRCUIT TYPE 2

Items		Symbol	Unit	Lower limit	Standard	Upper limit	Remarks
Termination resistance	Nominal value	Rc	Ω	60	-	65	Maximum deviation: ± 1 % Minimum power consumption: 110 mW
Pull-up, pull-down resistance	Nominal value	Rp	Ω	-	1300	-	Maximum deviation: ± 1 % Minimum power consumption: 110 mW
Termination capacitance	Nominal value	C	nF	-	4.7	-	Maximum deviation: ± 10 % Minimum Voltage 50V

[CANHW022a]: The power source of termination presented in Figure 5-3 shall be disconnected in ECU SLEEP State[1].

[1] Refer to [REF 2] for SLEEP state definition.

5.4. COMMON MODE CHOKE (CMC)

In case of HS-CAN, to determine whether the CMC is necessary, refer to specification requirements provided separately for each system.

[CANHW023a]: The CMC shall be within the range specified in Table 5-13.

Table 5-13: CMC CONSTANT

Item	Symbol	Unit	Lower limit	Standard	Upper limit	Remarks
Common mode inductance	L	µH	51	-	-	-

6. INITIAL SETTING OF COMMUNICATION IC

6.1. RESET PHASE AND CAN CONTROLLER CONFIGURATION

The reset phase is defined as the phase during which the ECU is initializing and configuring.

The CAN controller configuration consists of setting all the different CAN registers in order to guarantee a nominal communication behavior as defined in the different CAN specifications.

[CANHW024a]: During the reset phase, the ECU shall not disturb the CAN network communication by applying a dominant level.

[CANHW025a]: During the reset phase, the ECU shall not disturb the CAN network communication by disturbing recessive bus voltage.

6.2. RESET PHASE AND CAN CONTROLLER CONFIGURATION

[CANHW-HS026a]: Communication Speed shall be set to 500 kbps ± 0.3 % including drift due to ageing, temperature and all elements that can influence on the communication speed (e.g. PLL, frequency shift, etc...).

[CANHW-HS027a]: A quartz oscillator or a ceramic resonator shall be used.

[CANHW-HS0 28 a]: Communication Speed accuracy as specified in [CANHW-HS026a] shall be demonstrated in [REF 6] with detailed information on the Quartz drift, ceramic resonator, and PLL jitter over ageing and temperature.

[CANHW-HS029a]: In case of use of a ceramic resonator, the certificate of the resonator supplier shall be delivered and the value shall be reported on [REF 6].

<u>Note</u>: The measurement has to be made on the representative ECU, and not on evaluation board. Each modification of the PCB which may impact the pairing leads to a new certificate.

[CANHW-HS030a]: In case of use of a ceramic resonator, the use of a PLL to generate the CAN clock is forbidden.

[CANHW-HS031a]: One Sample Point (SP) per bit shall be used.

[CANHW-HS032a]: The CAN Controller Configuration shall be chosen in the Table 6-1.

[CANHW-HS033a]: The CAN Controller Configuration shall be reported in [REF 6], including the following parameters: BRP, SJW, Sample Point Position, Sampling Mode.

[CANHW-HS034a]: The CAN Controller Register values in Hexadecimal matching the CAN Controller configuration shall be reported in [REF 6], including the following parameters: BRP, SJW, Sample Point Position, Sampling Mode.

[CANHW-FD035a]: Communication Speed shall be set to 500 kbps ± 0.3 % for Arbitration Phase including drift due to ageing, temperature and all elements that can influence on the communication speed (e.g. PLL, frequency shift, etc...).

[CANHW-FD036a]: Communication Speed shall be set to 2Mbps + 0.3 % for Data Phase including drift due to ageing, temperature and all elements that can influence on the communication speed (e.g. PLL, frequency shift, etc…).

[CANHW-FD037a]: A quartz oscillator shall be used and the use of a ceramic resonator is forbidden.

[CANHW-FD038a]: Communication Speed accuracy as specified in [CANHW-FD35a] and [CANHW-FD36a] shall be demonstrated in [REF 6] with detailed information on the Quartz drift and PLL jitter over ageing and temperature.

[CANHW-FD039a]: One Sample Point (SP) per bit shall be used for Arbitration Phase.

[CANHW-FD040a]: A Secondary Sample Point (SSP) shall be used for Data Phase.

[CANHW-FD041a]: TDC (Time Delay Compensation) and TDM (Time Delay Measurement) shall be enabled for Data Phase when reading its own transmitted bits as depicted in Figure 6.

[CANHW-FD042a]: The CAN Controller Configuration for Arbitration Phase and Data Phase shall be chosen in the Table 6-2.

[CANHW-FD043a]: The CAN Controller Configuration for Arbitration Phase and Data Phase shall be reported in [REF 6], included the CAN Controller Registers setting the following parameters: BRP, SJW, Primary Sample Point Position, TDC, TDM, Secondary Sample Point Offset, Sampling Mode.

[CANHW-FD044a]: The CAN Controller Register values in Hexadecimal for Arbitration Phase and Data Phase shall be reported in [REF 6].

Table 6-1: CAN-HS Controller Configuration

Bit time NTq	SJW	Sample Point	Baud Rate Prescaler		
			1	2	3 or more
15 Tq	2 Tq	80.00 % (12 Tq)	Forbidden	OK	OK
16 Tq	2 Tq	81.25 % (13 Tq)	OK	OK	OK
17 Tq	2 Tq	82.35 % (14 Tq)	OK	OK	OK
18 Tq	2 Tq	83.33 % (15 Tq)	OK	OK	OK
19 Tq	2 Tq	84.21 % (16 Tq)	OK	OK	OK
20 Tq	2 Tq	85.00 % (17 Tq)	OK	OK	OK
21 Tq	3 Tq	80.95 % (17 Tq)	OK	OK	OK

Note: Baud Rate Prescaler is the number of controller clock counts that generates 1TQ.
This is an actual value. (Not register value)

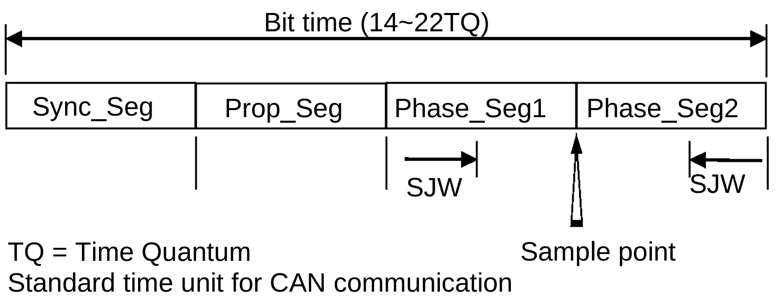


Figure 4: Structure of a Bit Time (CAN-HS)

Table 6-2: CAN-FD Controller Configuration

<table>
<tr><th rowspan="2">Configuration</th><th rowspan="2">CAN-FD Phase</th><th rowspan="2">Bit time NTq</th><th rowspan="2">SJW</th><th rowspan="2" colspan="2">Sample Point</th><th colspan="3">Baud Rate Prescaler</th></tr>
<tr><th>1</th><th>2</th><th>3 or more</th></tr>
<tr><td rowspan="3">Allowed Configuration #1</td><td>Arbitration Phase</td><td>80 Tq</td><td>16 Tq</td><td colspan="2">80% (64 Tq)</td><td>OK</td><td>OK</td><td>Forbidden</td></tr>
<tr><td rowspan="2">Data Phase</td><td rowspan="2">20 Tq</td><td rowspan="2">6 Tq</td><td>Primary</td><td>Secondary (Offset)</td><td rowspan="2">OK</td><td rowspan="2">OK</td><td rowspan="2">Forbidden</td></tr>
<tr><td>70% (14 Tq)</td><td>70% (14 Tq)</td></tr>
<tr><td rowspan="3">Allowed Configuration #2</td><td>Arbitration Phase</td><td>160 Tq</td><td>32 Tq</td><td colspan="2">80% (128 Tq)</td><td>OK</td><td>OK</td><td>Forbidden</td></tr>
<tr><td rowspan="2">Data Phase</td><td rowspan="2">40 Tq</td><td rowspan="2">12 Tq</td><td>Primary</td><td>Secondary (Offset)</td><td rowspan="2">OK</td><td rowspan="2">OK</td><td rowspan="2">Forbidden</td></tr>
<tr><td>70% (28 Tq)</td><td>70% (28 Tq)</td></tr>
</table>

This is an actual value. (Not register value)

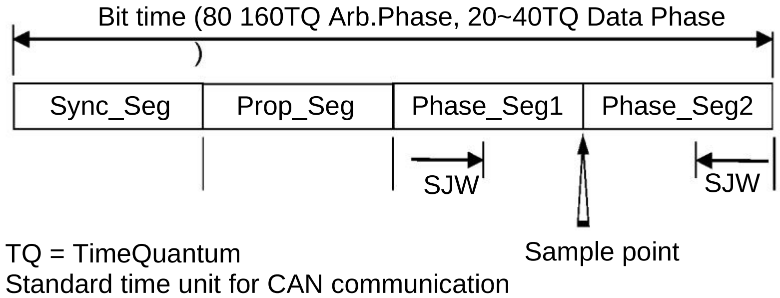


Figure 5: Structure of a Bit Time (CAN-FD)

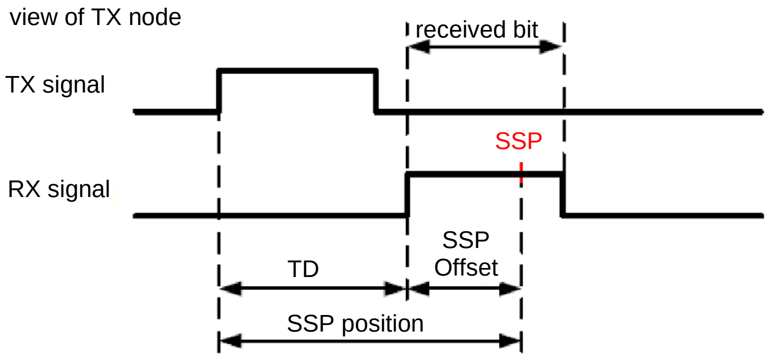


Figure 6: Time Delay Measurement and Secondary Sample Point

7. HARDWARE SPECIFICATION FOR WAKE UP BY CAN

[CANHW-045a]: In case of implementation of the wake up by CAN functionality, the filter time of the transceiver shall be chosen in the range [0.5 - 5µs] according to [REF 7].

Note: Refer to [REF 2] for information concerning the wake up by CAN strategies.

8. FAILURE MODE TOLERANCE

[CANHW046a]: When a failure (short-circuit, open-circuit...) occurs, the ECU shall behave as defined in the Table 8-1:

Table 8-1: FAILURE MODE TOLERANCE

<table>
<tr><th>Failure type</th><th colspan="2">Tolerated behaviour</th></tr>
<tr><td></td><td>Failure present</td><td>After the failure
(failure was present and disappeared)</td></tr>
<tr><td>Open circuit on CAN_L</td><td rowspan="4">The ECU shall not be destroyed.
The ECU shall not transmit "dominant" and shall not disturb the communication on the CAN_HS bus that may still exist between the other nodes.</td><td rowspan="10">The ECU shall not be destroyed.
Once the failure has disappeared, ECU shall recover nominal electrical characteristics, as specified in chapters 5.2 (1) to (8) of this document.</td></tr>
<tr><td>Open circuit on CAN_H</td></tr>
<tr><td>Ground loss on the ECU under test</td></tr>
<tr><td>Supply +Vbat loss on the ECU under test</td></tr>
<tr><td>Short-Circuit between CAN_H and +Vbat</td><td rowspan="4">The ECU shall not be destroyed.</td></tr>
<tr><td>Short-Circuit between CAN_L and +Vbat</td></tr>
<tr><td>Short-Circuit between CAN_H and Ground</td></tr>
<tr><td>Short-Circuit between CAN_H and CAN_L</td></tr>
<tr><td>Short-Circuit between CAN_L and Ground</td><td rowspan="2">The ECU shall not be destroyed.
The failure should not have any impact on the CAN communication. The ECU shall maintain its frame transmission and reception according to the ECU message set requirements.</td></tr>
<tr><td>Ground loss on another ECU</td></tr>
</table>

9. ANNEX A

(Normative)

10.ANNEX B

(Informative)

History of Modification (details)

DATE OF ISSUE

July 2001 - - - This issue originates from Draft NC 2000 0075 / - - -.

REVISIONS

April 2004 - - A

- Name change to “CAN High-Speed Hardware".
- Corrections and update.
- Renumbering of figures and tables.
- Addition of reference to EMC specification 36-00-808.
- Addition of a glossary.

§ 2.1.

- Addition of layout recommendations.
- Precision about noise filter and addition of EPCOS B82793 as authorized noise filter model.

§ 2.2.

- (4) TABLE 2-4.: Addition of title “ECU EQUIVALENT CIRCUIT” and of note about Cin and Cdiff.
- (6) TABLE 2-6.: Removal of 30ns lower limit for transition slope and addition of remark about EMC.

§ 2.3. (2) FIGURE 2-3.: Addition of note about power source disconnection in sleep state.
§ 3.1. Transfer of validation part (to § 7.9 in this version).
§ 3.2. CAN controller configurations (TABLE 3-2.) and simple sampling mode are now mandatory. Modification of TABLE 3-1. and 3-2..
§ 4.

- Transfer of Wake Up by CAN specification (§ 6. specification part of previous version).
- Change of “specific wake up frame” to “wake up frame”.

§ 5. Addition of specification of behavior during failure modes.
§ 6. Addition of tests validation process and necessary equipment description.
§ 7.

- Title change.
- 7.1.: VBATmin changed from 8 V to ECU specific.
- 7.2.1.: Correction of Rin_H formula (U = 0 V).
- 7.2.2.: Modification/correction of specification and calculation of RinECU.
- 7.2.2.: Test made not necessary if ECH has termination with bias.
- 7.8.: Transfer of Wake Up by CAN validation (§ 6. validation part of previous version).
- 7.9.: Transfer of reset phase validation (§ 3.1. validation part of previous version).

§ 8.

- Title change.
- Removal of short-circuit to - 3 V tests (5.11. and 5.12. in previous version).

§ ANNEX Modification of Sample point at 81.25 %
This issue originates from Draft NC 2004 0182 / ---.

March 2010 - - B

- Validation Test plan separated in a separated document
- Corrections and update.
- Renumbering of figures and tables.
- Use of requirement numbering.

$2.1.

- CMC and termination circuit pads are mandatory.

§2.2

- Removal of “ECU PROPAGATION DELAY TIME” specification and test.

$2.2.(4)

- Renaming of CAN lines resistances and capacitances
- Cin max. is now 260pF Renault Nissan Internal

		$2.3. • Addition of stub termination circuit (4.8kΩ) and precision of main termination Zt (120Ω). • Modification of minimum and maximum limits for terminations. **$2.4.** • Addition of minimum value allowed for CMC. **$3.1.** • Modification of CAN controller initialization requirement. **§3.2** • Addition of required information about oscillator system. • Ceramic resonator are now authorized if tolerance is lower than 0.27% and under certain conditions.
March 2016	- - C	• New Template • Update for CAN_FD • Clarification of Requirement Numbering • Clarification of clock Tolerance • Cin max. is now 50pF
March 2017	- - D	• Correction of Receiver Timing Asymmetry for 5Mbps Transceiver according to ISO11898-2 • Typography mistake corrections: correct reference to Table 6-1 and 6-2